

Haozhou Wang (王浩舟)

Project Assistant Professor, Laboratory of Field Phenomics,
Graduate School of Agricultural and Life Sciences, The University of Tokyo.

✉ haozhou-wang@outlook.com 🌐 haozhou.wang 📠 HowcanoeWang 📄 446 citations, h-index 10

Research Interests

- High-throughput plant 3D phenotyping.
- Digital twin virtual plant model and multi-sensory data fusion.
- Open-source agricultural phenotyping tool and dataset development.

Featured Projects and Publications

3DPotatoTwin dataset

[huggingface dataset](#) 📄

Dataset intended to help train and benchmark multi-sensory fusion and shape completion algorithms specifically for applications involving actual potato tuber harvesting.

- **Wang, H.**, Blok, P. M., Burridge, J., Jiang, T., Miyauchi, M., et al. (2025). 3DPotatoTwin: A Paired Potato Tuber Dataset for 3D Multi-Sensory Fusion. In: *Plant Phenomics* 7.4, p. 100123. DOI: [10.1016/j.plaphe.2025.100123](#) 📄 ;
- **Citations:** 0; **Reported by** [日本日経新聞](#) 📄

Broccoli harvest date prediction application

[github repo](#) 📄

A demonstrable application of aerial phenotyping technology to assist farmers in optimizing financial returns and minimizing food waste.

- **Wang, H.**, Li, T., et al. (2023). Drone-Based Harvest Data Prediction Can Reduce On-Farm Food Loss and Improve Farmer Income. In: *Plant Phenomics* 5, p. 0086. DOI: [10.34133/plantphenomics.0086](#) 📄 ;
- **Citations:** 21; **Reported by** [EurekAlert!](#) 📄, [日本日経新聞](#) 📄, and [新京报](#) 📄

EasyIDP intermediate data processing tool

[github repo](#) 📄

A handy tool for dealing with region of interest (ROI) on the image reconstruction (Metashape & Pix4D) outputs, mainly in agriculture applications.

- **Wang, H.**, Duan, Y., et al. (2021). EasyIDP: A Python Package for Intermediate Data Processing in UAV-based Plant Phenotyping. In: *Remote Sensing* 13.13, p. 2622. DOI: [10.3390/rs13132622](#) 📄.
- **Citations:** 40; **Github stars:** 47; **Tools Used:** Python, PyPi, Readthedocs;

UAV-HiRAP data processing platform

[uav-hirap.org](#) 📄

An open-source and web-based platform which provides service for image classification.

- **Wang, H.**, Han, D., et al. (2019). Landscape-Level Vegetation Classification and Fractional Woody and Herbaceous Vegetation Cover Estimation over the Dryland Ecosystems by Unmanned Aerial Vehicle Platform. In: *Agricultural and Forest Meteorology* 278, p. 107665. DOI: [10.1016/j.agrformet.2019.107665](#) 📄 ;
- **Citations:** 45; **Tools Used:** Python, Flask, Bootstrap, Nginx;

Professional Positions

Project Assistant Professor

The University of Tokyo

Tokyo, Japan

Feb. 2026 –present

- Greenhouse digital twin crop models.
- IoT sensor networks for greenhouse monitoring.
- Aerodynamic simulation and optimization for greenhouse structures.

Project Research Assistant

The University of Tokyo

Tokyo, Japan

Oct. 2023 –Jan. 2026

- A 3D paired potato tuber dataset for close-range multi-sensor data fusion.
- Aerial sensing system and software toolkit for detecting abnormal potatoes.
- Structural modeling and growth prediction framework for 3D virtual plants and digital twin.

Education

The University of Tokyo

Oct. 2020 –Sept. 2023

Doctor in Agricultural Science

Thesis title: Studies on 3D-based plant phenotyping by multi-scale data fusion.

The University of New Brunswick

Sept. 2017 –Dec. 2019

Master of Science in Forestry

Thesis title: Estimating Forest Attributes from Spherical Images.

The Nanjing Forestry University

Sept. 2013 –Jun. 2017

Bachelor of Science in Ecology

Thesis title: Extracting DBH Measurements from RGB Photo Images.

Publications

Book Chapters (1 entries)

Wang, H., Guo, W. (2024). EasyIDP V2.0: An Intermediate Data Processing Package for Photogrammetry-Based Plant Phenotyping. In: *Harnessing Data Science for Sustainable Agriculture and Natural Resource Management*. Ed. by M. S. Raval, S. Chaudhary, J. Adinarayana, and W. Guo. Vol. 161. Singapore: Springer Nature Singapore, pp. 149–172. DOI: [10.1007/978-981-97-7762-4_7](https://doi.org/10.1007/978-981-97-7762-4_7) .

Patent (1 entries)

Wang, F., Han, D., **Wang, H.**, Lu, Q., Pan, X. (2017). Landscape-Scale Vegetation Coverage Calculation Method and System Based on Unmanned Aerial Vehicle (一种基于无人机的景观尺度植被覆盖度的计算方法及系统). Pat. CN106403904A (CN). Chinese Academy of Forestry.

Journal articles (24 entries)

Drofova, I., **Wang, H.**, Guo, W., Katsuhama, N., Burrridge, J., Blok, P. M., Adamek, M. (2026). Color Vision in Digital Twin Creation Using Photogrammetry in Sustainable Agriculture 4.0. In: *Sustainability* 18.5, p. 2160. DOI: [10.3390/su18052160](https://doi.org/10.3390/su18052160) .

Wang, H., Guo, W., et al. (2026). Multi-Scale Spatial-Temporal Remote Sensing Fusion for Phenology Identification in Rice Germplasm Resources. In: *Plant Phenomics* 8.3, p. 100222. DOI: [10.1016/j.plaphe.2026.100222](https://doi.org/10.1016/j.plaphe.2026.100222) .

Blok, P. M., Magistri, F., Stachniss, C., **Wang, H.**, Burrridge, J., Guo, W. (2025). High-Throughput 3D Shape Completion of Potato Tubers on a Harvester. In: *Computers and Electronics in Agriculture* 228, p. 109673. DOI: [10.1016/j.compag.2024.109673](https://doi.org/10.1016/j.compag.2024.109673) .

Li, H., Liu, H., Wang, W., **Wang, H.**, Yu, Q., Qian, J., Wu, W., Shi, Y., Geng, C. (2025). A Point-Supervised Algorithm with Multiscale Semantic Enhancement for Counting Multiple Crop Plants from Aerial Imagery. In: *Computers and Electronics in Agriculture* 234, p. 110289. DOI: [10.1016/j.compag.2025.110289](https://doi.org/10.1016/j.compag.2025.110289) .

Liu, H., Li, H., **Wang, H.**, Liu, C., Qian, J., Wang, Z., Geng, C. (2025). Improved Detection and Location of Small Crop Organs by Fusing UAV Orthophoto Maps and Raw Images. In: *Remote Sensing* 17.5, p. 906. DOI: [10.3390/rs17050906](https://doi.org/10.3390/rs17050906) .

Wang, H., Blok, P. M., Burrridge, J., Jiang, T., Miyauchi, M., Miyamoto, K., Tanaka, K., Guo, W. (2025). 3DPotatoTwin: A Paired Potato Tuber Dataset for 3D Multi-Sensory Fusion. In: *Plant Phenomics* 7.4, p. 100123. DOI: [10.1016/j.plaphe.2025.100123](https://doi.org/10.1016/j.plaphe.2025.100123) .

Wang, Z. et al. (2025). The Global Wheat Full Semantic Organ Segmentation (GWFSS) Dataset. In: *Plant Phenomics* 7.3, p. 100084. DOI: [10.1016/j.plaphe.2025.100084](https://doi.org/10.1016/j.plaphe.2025.100084) .

Zhang, W., Zheng, C., Wang, C., Blok, P. M., **Wang, H.**, Guo, W. (2025). GrapeCPNet: A Self-Supervised Point Cloud Completion Network for 3D Phenotyping of Grape Bunches. In: *Computers and Electronics in Agriculture* 237, p. 110595. DOI: [10.1016/j.compag.2025.110595](https://doi.org/10.1016/j.compag.2025.110595) .

Zhou, J. et al. (2025). Global Rice Multiclass Segmentation Dataset (RiceSEG): Comprehensive and Diverse High-Resolution RGB-annotated Images for the Development and Benchmarking of Rice Segmentation Algorithms. In: *Plant Phenomics* 7.3, p. 100099. DOI: [10.1016/j.plaphe.2025.100099](https://doi.org/10.1016/j.plaphe.2025.100099) .

Zhang, W., Peng, X., Bai, T., **Wang, H.**, Takata, D., Guo, W. (2024). A UAV-Based Single-Lens Stereoscopic Photography Method for Phenotyping the Architecture Traits of Orchard Trees. In: *Remote Sensing* 16.9, p. 1570. DOI: [10.3390/rs16091570](https://doi.org/10.3390/rs16091570) .

Drofova, I., Guo, W., **Wang, H.**, Adamek, M. (2023). Use of Scanning Devices for Object 3D Reconstruction by Photogrammetry and Visualization in Virtual Reality. In: *Bulletin of Electrical Engineering and Informatics* 12.2, pp. 868–881. DOI: [10.11591/eei.v12i2.4584](https://doi.org/10.11591/eei.v12i2.4584) .

- Wang, H.**, Li, T., Nishida, E., Kato, Y., Fukano, Y., Guo, W. (2023). Drone-Based Harvest Data Prediction Can Reduce On-Farm Food Loss and Improve Farmer Income. In: *Plant Phenomics* 5, p. 0086. DOI: [10.34133/plantphenomics.0086](https://doi.org/10.34133/plantphenomics.0086) .
- Zhang, W., Peng, X., Cui, G., **Wang, H.**, Takata, D., Guo, W. (2023). Tree Branch Skeleton Extraction from Drone-Based Photogrammetric Point Cloud. In: *Drones* 7.2, p. 65. DOI: [10.3390/drones7020065](https://doi.org/10.3390/drones7020065) .
- Dai, X., Ducey, M. J., **Wang, H.**, Yang, T.-R., Hsu, Y.-H., Ogilvie, J., Kershaw, J. A. (2021). Biomass Estimates Derived from Sector Subsampling of 360 Spherical Images. In: *Forestry: An International Journal of Forest Research* 94.4, pp. 565–575. DOI: [10.1093/forestry/cpab023](https://doi.org/10.1093/forestry/cpab023) .
- Dai, X., Ducey, M. J., Kershaw, J. A., **Wang, H.** (2021). Sector Subsampling for Basal Area Ratio Estimation: An Alternative to Big BAF Sampling. In: *Canadian Journal of Forest Research* 51.8, pp. 1–9. DOI: [10.1139/cjfr-2020-0496](https://doi.org/10.1139/cjfr-2020-0496) .
- David, E. et al. (2021). Global Wheat Head Detection 2021: An Improved Dataset for Benchmarking Wheat Head Detection Methods. In: *Plant Phenomics* 2021, p. 9846158. DOI: [10.34133/2021/9846158](https://doi.org/10.34133/2021/9846158) .
- Feldman, A., **Wang, H.**, Fukano, Y., Kato, Y., Ninomiya, S., Guo, W. (2021). EasyDCP: An Affordable, High-Throughput Tool to Measure Plant Phenotypic Traits in 3D. In: *Methods in Ecology and Evolution* 12.9, pp. 1679–1686. DOI: [10.1111/2041-210X.13645](https://doi.org/10.1111/2041-210X.13645) .
- Hsu, Y.-H., Kershaw, J. A., Ducey, M. J., Yang, T.-R., **Wang, H.** (2021). Sampling with Probability Proportional to Prediction (3P Sampling) Using Covariates Derived from Spherical Images. In: *Canadian Journal of Forest Research* 51.8, pp. 1140–1147. DOI: [10.1139/cjfr-2020-0498](https://doi.org/10.1139/cjfr-2020-0498) .
- Wang, H.**, Duan, Y., Shi, Y., Kato, Y., Ninomiya, S., Guo, W. (2021). EasyIDP: A Python Package for Intermediate Data Processing in UAV-based Plant Phenotyping. In: *Remote Sensing* 13.13, p. 2622. DOI: [10.3390/rs13132622](https://doi.org/10.3390/rs13132622) .
- Wang, H.**, Yang, T.-R., Waldey, J., Kershaw, J. A. (2021). Estimating Individual Tree Heights and DBHs from Vertically Displaced Spherical Image Pairs. In: *Mathematical and Computational Forestry & Natural-Resource Sciences* 13.1 (1), pp. 1–14.
- Zhao, L., Guo, W., Wang, J., **Wang, H.**, Duan, Y., Wang, C., Wu, W., Shi, Y. (2021). An Efficient Method for Estimating Wheat Heading Dates Using UAV Images. In: *Remote Sensing* 13.16, p. 3067. DOI: [10.3390/rs13163067](https://doi.org/10.3390/rs13163067) .
- Wang, H.**, Kershaw, J. A., Yang, T.-R., Hsu, Y.-H., Ma, X., Chen, Y. (2020). An Integrated System for Estimating Forest Basal Area from Spherical Images. In: *Mathematical and Computational Forestry & Natural-Resource Sciences* 12.1, pp. 0–14.
- Wang, H.**, Han, D., Mu, Y., Jiang, L., Yao, X., Bai, Y., Lu, Q., Wang, F. (2019). Landscape-Level Vegetation Classification and Fractional Woody and Herbaceous Vegetation Cover Estimation over the Dryland Ecosystems by Unmanned Aerial Vehicle Platform. In: *Agricultural and Forest Meteorology* 278, p. 107665. DOI: [10.1016/j.agrformet.2019.107665](https://doi.org/10.1016/j.agrformet.2019.107665) .
- Han, D., **Wang, H.**, Zheng, B., Wang, F. (2018). Vegetation Type Classification and Fractional Vegetation Coverage Estimation for an Open Elm (*Ulmus Pumila*) Woodland Ecosystem during a Growing Season Based on an Unmanned Aerial Vehicle Platform Coupled with Decision Tree Algorithms. In: *Acta Ecologica Sinica* 38.18, pp. 6655–6663. DOI: [10.5846/stxb201803300694](https://doi.org/10.5846/stxb201803300694) .

Conference proceedings (18 entries)

- Xiang, S., Blok, P. M., Burrridge, J., **Wang, H.**, Guo, W. (2026). “DODA: Adapting Object Detectors to Dynamic Agricultural Environments in Real-Time with Diffusion”. In: *2026 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*. 2026 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV). Tucson, AZ, USA: IEEE, pp. 4797–4807. DOI: [10.1109/WACV61042.2026.00466](https://doi.org/10.1109/WACV61042.2026.00466) .
- Wang, H.**, Gong, Z., Liu, H., Li, H., Guo, W., Su, B., Shi, Y. (2025). Camera Align and Folder Drag Are All You Need: Rapid Crop Lodging Aerial Assessment without Segmentation. Oral. Tokyo, Japan: Seventh International Workshop on Machine Learning for Cyber-Agricultural Systems (MLCAS2025).
- Wang, H.**, James, C., Chapman, S. C., Guo, W. (2025). StagGCP: A Metashape Plugin for Using STag as Robust Ground Control Points for In-Field Agricultural 3D Reconstruction. Oral. Kyoto, Japan: Annual Conference of the Japanese Society of Agricultural Informatics 2025 (農業情報学会 JSAI 2025 年次大会).
- Wang, H.**, Blok, P. M., Burrridge, J., Jiang, T., Guo, W. (2024). 3DPotatoTwin: Paired 3D Dataset of Potato Tubers for Plant Phenotyping Applications. Poster. Jeju International Convention Center (ICC JEJU), Jeju, Korea: The 6th CIGR International Conference 2024 (CIGR2024).
- Wang, H.**, Inoshishi, S., Shimizu, M., Kato, T., Guo, W. (2024). Drone-Based Multi-spectral Pipeline for Detecting Abnormal Potato Strains in the Field. Oral. Tsukuba International Congress Center, Tsukuba, Japan: The 14th Asia-Pacific Federation for Information Technology in Agriculture 2024 (APFITA2024).
- Wang, H.**, Tang, L., Nishida, E., Fukano, Y., Kato, Y., Guo, W. (2023a). Virtual Broccoli Farmland by Fusing Close-Range and Aerial Phenotyping. Oral. Sarabetsu Village, Hokkaido, Japan: Fifth International Workshop on Machine Learning for Cyber-Agricultural Systems (MLCAS2023).

- Wang, H.**, Tang, L., Nishida, E., Fukano, Y., Kato, Y., Guo, W. (2023b). Virtual Broccoli Farmland Implementation by Drone-Based Phenotyping and Cross-Scale Data Fusion. Oral. Guangzhou, China: The 10th International Horticulture Research Conference (IHRC2023).
- Wang, H.**, Kato, Y., Guo, W. (2022). Procedural Geometric Modeling for Plant Phenomics by Blender: Case Study of Maize. Oral. Kyoto, Japan: Annual Conference of the Japanese Society of Agricultural Informatics 2022 (農業情報学会 2022 年次大会, JSAI2022).
- Wang, H.**, Tang, L., Nishida, E., Fukano, Y., Kato, Y., Guo, W. (2022). Estimate Optimal Harvest Time by Cross-scale Assimilated Digital Broccoli Farmland. Poster. Wageningen, Netherlands.: 7th International Plant Phenotyping Symposium: "Plant Phenotyping for a Sustainable Future".
- Wang, H.**, Tang, L., Nishida, E., Fukano, Y., Kato, Y., Guo, W. (2021). Cost-Efficient Broccoli Head Phenotyping Using Aerial Imagery and SfM-based Weakly Supervised Learning. Nanjing, Jiangsu, China: The 8th International Horticulture Research Conference.
- Wang, H.**, Yoichiro, K., Guo, W. (2021a). EasyIDP: A Python Package for Intermediate Data Processing in UAV Based Plant Phenotyping. Zoom online, Tokyo, Japan.: Annual Conference of the Japanese Society of Agricultural Informatics 2021 (農業情報学会 2021 年次大会, JSAI2021).
- Wang, H.**, Yoichiro, K., Guo, W. (2021b). EasyIDP: A Python Package for Intermediate Data Processing in UAV Based Plant Phenotyping. Zoom online, Tokyo, Japan.: The 1th Research Meeting of Society of Trans-disciplinary Plant Sciences (超分野植物科学研究会第 1 回研究集会, TDPS2021).
- Feldman, A., **Wang, H.**, Fukano, Y., Kato, Y., Ninomiya, S., Guo, W. (2020). Affordable High-Throughput Processing of Handheld Camera Images of Container Plants to Phenotypic Data. Tucson Convention Center, Tucson, Arizona, USA.: Phenome 2020, p. 1.
- Feldman, A., **Wang, H.**, Fukano, Y., Guo, W. (2019). Affordable High-Throughput Processing of Multi-Scale Images to Phenotypic Data. Nanjing Dongjiao State Guesthouse, Nanjing, Jiangsu, China.: The 6th International Plant Phenotyping Symposium, p. 1.
- Wang, H.**, Kershaw, J. A. (2019). Estimating Forest Attributes from Spherical Images. Oral, Poster. Kamloops Hotel and Conf. Center, Kamloops, British Columbia, Canada: The Western Mensurationists 2019 Annual Meeting.
- Wang, H.**, Kershaw, J. A. (2018). Measuring Plant Area Index (PAI) from Panorama Photo Images. Oral. Wu Conference Center, Fredericton, New Brunswick, Canada.: The 25th Annual UNB Graduate Research Conference (GRC).
- Wang, H.**, Kershaw, J. A. (2017). Extracting DBH Measurements from RGB Photo Images. Oral. New York, U.S: The Northeastern Mensurationists 2017 Annual Meeting.
- Wang, H.**, Wang, F., Yao, X., Mu, Y., Bai, Y., Lu, Q. (2017). UAV-HiRAP: A Novel Method to Improve Landscape-Level Vegetation Classification and Coverage Fraction Estimation with Unmanned Aerial Vehicle Platform. Oral. Beijing, China: The 12th International Congress of Ecological (INTECOL).

Awards

The 6th CIGR international conference young researcher travel award.	<i>May 2024</i>
The 12th JSAI (農業情報学会年次大会) young researcher innovation award.	<i>May 2021</i>
The third prize of the 8th Liang Xi youth paper award (梁希青年论文奖).	<i>Nov 2020</i>
The first place for oral presentation on 25th UNB GSA Conference.	<i>May 2018</i>